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United States Patent	12413187
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hua; Qi et al.

Doherty amplifiers

Abstract

A Doherty amplifier includes first and second input terminals, first and second amplifiers, and an output combiner circuit. The first amplifier includes a first amplifier input coupled to the first input terminal, and a first amplifier output. The second amplifier includes a second amplifier input coupled to the second input terminal, and a second amplifier output. The output combiner circuit is coupled between the first amplifier output, the second amplifier output, and a final summing node. The output combiner circuit includes a first inductive element, a first capacitor integrated within an integrated passive device (IPD), and a second inductive element. The first inductive element is coupled between the first amplifier output and a first terminal of the first capacitor, and the second inductive element is coupled between a combining node and the first terminal of the first capacitor. A second terminal of the first capacitor is coupled to ground.

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Appl. No.: 18/052586

Filed: November 04, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240063756 A1	Feb. 22, 2024

Foreign Application Priority Data

CN	202210996208.8	Aug. 18, 2022
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Publication Classification

Int. Cl.: **H03F1/02** (20060101); **H01L23/00** (20060101); **H01L23/66** (20060101); **H01L25/16** (20230101); **H03F1/56** (20060101); **H03F3/195** (20060101)

U.S. Cl.:

CPC **H03F1/0288** (20130101); **H01L23/66** (20130101); **H01L24/48** (20130101); **H01L25/16** (20130101); **H03F1/565** (20130101); **H03F3/195** (20130101); H01L2223/6611 (20130101); H01L2223/6655 (20130101); H01L2223/6672 (20130101); H01L2224/48011 (20130101); H01L2224/48091 (20130101); H01L2224/48195 (20130101); H01L2924/19011 (20130101); H01L2924/19041 (20130101); H01L2924/19042 (20130101); H01L2924/19043 (20130101); H01L2924/19105 (20130101); H01L2924/2027 (20130101); H01L2924/30111 (20130101); H03F2200/391 (20130101)

Field of Classification Search

CPC: H03F (5/00); H03F (3/45475); H03F (3/44); H03F (3/28); H03F (3/602); H03F (7/02); H03F (3/45085); H03F (3/26); H03F (3/88); H03F (1/14); H03F (3/21); H03F (2200/372); H03F (3/68); H03G (1/0023); H03G (1/00); H01L (2924/3011)

USPC: 330/3; 330/69; 330/74; 330/116; 330/117; 330/124; 330/134; 330/252; 330/254; 330/262; 330/275; 330/284; 330/292; 330/295; 330/301; 330/307; 330/124R; 375/296-297; 455/114.3

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6262629	12/2000	Stengel et al.	N/A	N/A
8022760	12/2010	Gajadharsing et al.	N/A	N/A
8228123	12/2011	Blednov	N/A	N/A
8653889	12/2013	Acimovic	330/124R	H03F 3/245
9509252	12/2015	Moronval et al.	N/A	N/A
10038407	12/2017	Moronval et al.	N/A	N/A
2005/0231278	12/2004	Blednov	N/A	N/A
2009/0174482	12/2008	Blednov	N/A	N/A
2014/0062601	12/2013	Acimovic et al.	N/A	N/A
2015/0145601	12/2014	Moronval	330/295	H03F 1/07
2020/0186097	12/2019	Hue	N/A	H03F 3/604

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110417357	12/2018	CN	H03F 1/0288
2005043747	12/2004	WO	N/A

OTHER PUBLICATIONS

Blednov, I; “Wideband 3 Way Doherty RFIC with 12 dB Back-Off Power Range”; Proceedings of the 11th European Microwave Integrated Circuits Conference; London, UK; 4 pages (Oct. 2016). cited by applicant

Cygan Lawrence F. “A High Efficiency Linear Power Amplifier for Portable Communications Applications”, IEEE Compound Semiconductor Integrated Circuit Symposium, 2005. CSIC '05., 2005, 5 pages. cited by applicant

Maroldt, Stephan et al; “3.5-GHz Ultra-Compact GaN Class-E Integrated Doherty MMIC PA for 5G massive-MIMO Base Station Applications”; Proceedings of the 12th European Microwave Integrated Circuits Conference, Nuremberg DE; 4 pages (Oct. 9-10, 2017). cited by applicant

Unknown; “A Compact 60 W MMIC Amplifier based on a Novel 3-Way 1:2:1 Doherty Architecture with Best-in-Class Efficiency for Small Cells”; IEEE MTT-S Intl Microwave Symposium; 4 pages (2017). cited by applicant

U.S. Appl. No. 17/929,527, filed Sep. 2, 2022, not yet publish, 38 pages—US counterpart of CN Application 202111038476.0. cited by applicant

Zhang, Hao et al; “A 50W Highly Linear 3-Way Integrated Wideband Doherty PA for Small-Cell Application”; Proceedings of the 48th European Microwave Conference; 4 pages (2018). cited by applicant

Primary Examiner: Rahman; Hafizur

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority under 35 U.S.C. § 119 to Chinese patent application no. 202210996208.8, filed on 18 Aug. 2022, the contents of which are incorporated by reference herein.

BACKGROUND

(2) The present application is generally directed to amplifiers, and more particularly to Doherty amplifiers and devices used in the Doherty amplifiers.

(3) With the development of the 5G communication market, energy savings at the low traffic state become more and more important. Therefore, more and more telecom operators require high efficiency at the low traffic state (e.g., typically half output power of normal operations, or even more less), meanwhile requiring the same or even higher efficiency at normal operational output power.

(4) 2-way and 3-way Doherty amplifier circuits are often considered for use in 5G base stations, because they tend to have relatively high efficiency. The 3-way Doherty amplifier circuit usually has a complex output combiner, together with output matching networks for each individual path. Accordingly, it takes a large amount of printed circuit board (PCB) room to accommodate them. In the macro base station, the circuit size may be not a big problem because it does not have many channels (e.g., 2 channels or 4 channels, typically). However, a 5G Massive MIMO communication system requires multiple channels of high efficiency power amplifiers in each 5G base station. Either 32T (32 channels) or 64T (64 channels) is a common requirement, and therefore implementing Doherty amplifiers in Massive MIMO systems brings many challenges, especially on the circuit size, as well as the cost.

(5) Therefore, there remains a need for Doherty amplifiers which may have the advantages of size reduction, cost savings and high efficiency, both at deep back-off and normal output power levels.

SUMMARY

(6) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to be relied on to identify important features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(7) According to one aspect of the application, it provides a Doherty amplifier device, comprising: a first input terminal; a second input terminal; a third input terminal; an integrated output combiner circuit comprising a first output network, a second output network and a third output network, an intermediate summing node, and a final summing node; a main amplifier including a first amplifier input coupled to the first input terminal, and a first amplifier output coupled to the first output network; a first peaking amplifier including a second amplifier input coupled to the second input terminal, and a second amplifier output coupled to the second output network through the intermediate summing node; and a second peaking amplifier including a third amplifier input coupled to the third input terminal, and a third amplifier output coupled to the third output network; wherein the first output network is coupled between the first amplifier output and the final summing node, the second output network coupled between the second amplifier output and the final summing node, and the third output network coupled between the third amplifier output and the intermediate summing node; wherein the first output network imparts a phase delay of 90 degrees between an input of the first output network and an output of the first output network and provide a first impedance, the second output network imparts a phase delay of 90 degrees between an input of the second output network and an output of the second output network and provide a second impedance, and the third output network imparts a phase delay of 90 degrees between an input of the third output network and an output of the third output network and provide a third impedance; wherein the first output network comprises a first inductive element coupled between the first amplifier output and a first intermediate node, a first capacitor integrated within a first integrated passive device (IPD) and coupled between the first intermediate node and ground, and a second inductive element coupled between the first intermediate node and the final summing node.

(8) According to the second aspect of the application, it provides a Doherty amplifier, comprising: a first input; a second input; a third input; an output; an output combiner circuit with three combining network inputs, multiple summing nodes, and a combining network output coupled to the output directly; a main amplifier including a first amplifier input coupled to the first input, and a first amplifier output coupled to a first combining network input of the three combining network inputs; a first peaking amplifier including a second amplifier input coupled to the second input, and a second amplifier output coupled to a second combining network input of the three combining network inputs; and a second peaking amplifier including a third amplifier input coupled to the third input, and a third amplifier output coupled to a third combining network input of the three combining network inputs; wherein the phase shift between the first amplifier input and the second amplifier input is 0° , and the phase shift between the second amplifier input and the third amplifier input is 90° ; wherein the multiple summing nodes of the output combining network include a final summing node and an intermediate summing node, and wherein the intermediate summing node is coupled to the second combining network input directly; wherein the output combiner circuit comprises a first output network between the first amplifier output and the combining network output, a second output network between the intermediate summing node and the final summing node and a third output network between the third amplifier output and the intermediate summing node; wherein the first output network imparts a phase delay of 90 degrees between an input of the first output network and an output of the first output network and provide a first impedance, the second output network imparts a phase delay of 90 degrees between an input of the second output network and an output of the second output network and provides a second impedance, and the third output network imparts a phase delay of 90 degrees between an input of the third output network and an output of the third output network and provides a third impedance, and the first

output network comprises a first inductive element coupled between the first amplifier output and a first intermediate node, a first capacitor integrated within a first integrated passive device (IPD) and coupled between the first intermediate node and ground, and a second inductive element coupled between the first intermediate node and the final summing node.

(9) According to the third aspect of the application, it provides a Doherty amplifier device, comprising: a first input terminal; a second input terminal; an integrated output combiner circuit comprising a first output network and a second output network; a main amplifier including a first amplifier input coupled to the first input terminal, and a first amplifier output coupled to the first output network; and a peaking amplifier including a second amplifier input coupled to the second input terminal, and a second amplifier output coupled to the second output network, wherein the first output network comprises a first inductive element coupled between the first amplifier output and a first terminal of a first capacitor that is integrated within a first integrated passive device (IPD), a second inductive element coupled between a final summing node and the first terminal of the first capacitor, and wherein a second terminal of the first capacitor is coupled to ground, and wherein the second output network comprises a third inductive element coupled between the second amplifier output and a first terminal of a second capacitor that is integrated within a second IPD, a fourth inductive element coupled between the final summing node and the first terminal of the second capacitor, and wherein a second terminal of the second capacitor is coupled to ground.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present application can be understood in detail, a more particular description of the application, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this application and are therefore not to be considered limiting of its scope, for the application may admit to other equally effective embodiments. The drawings are for facilitating an understanding of the application and thus are not necessarily drawn to scale. Advantages of the subject matter claimed will become apparent to those skilled in the art upon reading this description in conjunction with the accompanying drawings, in which like reference numerals have been used to designate like elements, and in which:

- (2) FIG. 1 illustrates a simplified schematic diagram of a Doherty amplifier with transmission lines;
- (3) FIG. 2A illustrates a 3-way Doherty amplifier in accordance with an embodiment;
- (4) FIG. 2B illustrates a 3-way Doherty amplifier using C-L-C-L-C circuits in accordance with an embodiment;
- (5) FIG. 2C illustrates a quasi quarter-wave transmission line formed by a C-L-C-L-C circuit;
- (6) FIG. 3 illustrates the physical layout of a 3-way Doherty amplifier device in accordance with an embodiment;
- (7) FIG. 4 illustrates a schematic perspective diagram of a packaged amplifier device corresponding to the Doherty amplifier of FIG. 3 in accordance with an embodiment;
- (8) FIG. 5 illustrates a simplified schematic diagram of a Doherty amplifier with 2nd harmonic control;
- (9) FIG. 6 illustrates the physical layout of an amplifier device corresponding to the Doherty amplifier of FIG. 5 in accordance with an embodiment;
- (10) FIG. 7 illustrates a Smith chart illustrating second harmonic load-pull data;
- (11) FIG. 8 illustrates a Smith chart showing an example load impedance for a Doherty amplifier without 2nd harmonic control;
- (12) FIG. 9 illustrates a Smith chart showing an example load impedance for a Doherty amplifier

with 2nd harmonic control;

(13) FIG. **10** illustrates a 2-way Doherty amplifier in accordance with an embodiment;

(14) FIG. **11** illustrates the physical layout of a 2-way Doherty amplifier device in accordance with an embodiment.

DETAILED DESCRIPTION

(15) FIG. **1** is a simplified schematic diagram of a three-way Doherty amplifier **100**. Amplifier **100** includes a single input terminal **102**, an final summing node **104**, a power splitter **110**, a first amplifier path **130**, a second amplifier path **140**, a third amplifier path **150**, and a combining terminal **160**. A load **106** (e.g., an antenna) may be coupled to the combining terminal **160** through an impedance transformer **108**. The impedance transformer **108** may impart a 90 degree phase delay to the output RF signal before it is supplied to the load **106**.

(16) Power splitter **110** is configured to divide the input power of an input signal **102** received at power splitter input **112** into carrier and peaking portions of the input signal. The carrier input signal is provided to the first amplifier path **130** at power splitter output **114**, a first peaking input signal is provided to the second amplifier path **140** at power splitter output **116**, and a second peaking input signal is provided to the third amplifier path **150** at power splitter output **118**. During operation in a full-power mode when all of the carrier amplifier **136**, second amplifiers **146** and third amplifier **156** are supplying current to the load **106**, the power splitter **110** divides the input signal power among the amplifier paths **130**, **140** and **150**. The power splitter **110** may divide the power unequally. For example, the power splitter **110** may divide the power 1:1.5:1.5, such that the Doherty amplifier **100** has a 1:1.5:1.5 configuration.

(17) Essentially, the power splitter **110** divides an input RF signal supplied at the input terminal **102**, and the divided signals are separately amplified along the first, second and third amplifier paths **130**, **140**, **150**. The amplified signals are then combined in phase at the combining terminal **160**. It is important that phase coherency between the first, second and third amplifier paths **130**, **140**, **150** is maintained across a frequency band of interest to ensure that the amplified carrier and peaking signals arrive in phase at the combining terminal **160**, and thus to ensure proper Doherty amplifier operation.

(18) Each of the first amplifier **136**, the second amplifier **146** and the third amplifier **156** includes one or more single stage or multiple-stage power transistor integrated circuits (ICs) **138**, **148**, **158** for amplifying an RF signal conducted through the amplifier **136**, **146**, **156**. These power transistor ICs may be implemented, for example, using silicon-based field effect transistors (FETs) (e.g., laterally diffused metal oxide semiconductor FETs, or LDMOS FETs), gallium nitride (GaN)-based FETs (e.g., high electron mobility transistors), or other types of power transistors. Although the carrier and peaking power transistor ICs may be of equal size (e.g., in a symmetric Doherty configuration), the carrier and peaking power transistor ICs may have unequal sizes, as well (e.g., in various asymmetric Doherty configurations). In an asymmetric Doherty configuration, the peaking power transistor ICs typically are larger than the carrier power transistor IC(s) by some multiplier.

(19) During operation of Doherty amplifier **100**, first amplifier stage **136** is biased to operate in class AB mode, and second and third amplifier stages **146**, **156** are biased to operate in class C mode. More specifically, the transistor arrangement of first amplifier stage **136** is biased to provide a conduction angle between 180 and 360 degrees. Conversely, the transistor arrangement of the second and third amplifier stages **146**, **156** are biased to provide a conduction angle less than 180 degrees.

(20) At low power levels, where the power of the input signal at terminal **102** is lower than the turn-on threshold level of second amplifier **146**, the amplifier **100** operates in a low-power (or back-off) mode in which the first amplifier **136** is the only amplifier supplying current to the load **106**. When the power of the input signal exceeds a threshold level of the second amplifier **146**, the amplifier **100** operates in a high-power mode in which the first amplifier **136**, the second amplifier

146 and the third amplifier **156** together supply current to the load **106**. At this point, the second amplifier and third amplifier **146, 156** provides active load modulation at combining terminal **160**, allowing the current of the first amplifier **136** to continue to increase linearly.

(21) Input impedance matching network **134** (input MNc) may be implemented at the input of the first amplifier **136**. Similarly, input impedance matching network **144** (input MNp1) may be implemented at the input of the second amplifier **146** and input impedance matching network **154** (input MNp2) may be implemented at the input of the third amplifier **156**. In each case, the matching networks **134, 144, 154** may be used to incrementally increase the circuit impedance toward the load impedance. In addition, the first amplifier **136**, the second amplifier **146** and the third amplifier **156** may have additional pre-matching input and/or output impedance matching networks (not illustrated) that are either integrated with the power transistor dies, or integrated within the power transistor die packages.

(22) Doherty amplifier **100** has a “inverted” load network configuration. In the inverted configuration, the input circuit is configured so that an input signal supplied to the first amplifier **136** and the second amplifier **146** are delayed by 90 degrees with respect to the input signal supplied to the third amplifier **156** at the center frequency of operation, f_0 , of the amplifier **100**. To ensure that the carrier and peaking input RF signals fundamental to proper Doherty operation, a first phase delay element **131** and a second phase delay element **132** are provided, in the input paths to the first amplifier **136** and the second amplifier **146** respectively, which apply about 90 degrees of phase delay to the carrier input signal and the first peaking input signal. For example, each of the phase delay element **131** and the phase delay element **132** may be a quarter wave transmission line, another suitable type of delay element with an electrical length of about 90 degrees.

(23) Delay elements **151** and **152** are implemented between the outputs of the first and second amplifiers **136, 146** and the combining terminal **160**. To compensate for the resulting 90 degree phase delay difference between second and third amplifier paths **140, 150** at the inputs of amplifiers **146, 156** (i.e., to ensure that the amplified signals arrive in phase at the combining terminal **160**), the output circuit is configured to apply about a 90 degree phase delay to the signal between the output of second amplifier **146** and the output of third amplifier **156**. This is achieved through an additional delay element **153**. Delay element **151** is configured to apply about a 90 degree phase delay to the signal between the output of first amplifier **136** and the combining terminal **160**, and delay element **152** is configured to apply about a 90 degree phase delay to the signal between the output of second amplifier **146** and the combining terminal **160**.

(24) In a printed circuit board (PCB) implementation of amplifier **100**, delay elements **151, 152, 153** are implemented as quarter-wave transmission lines or “quasi” transmission lines (e.g., a transmission line coupled to an amplifier with bond wires). However, the transmission line types of delay elements **151-153** are not readily able to be integrated in discrete power amplifier devices with certain common package types (e.g. dual flat no-leads, DFN, packages), because such types of packages do not include internal substrates that are suitable for supporting transmission lines.

(25) FIG. 2A is a simplified schematic diagram of a 3-way Doherty amplifier **200** in accordance with an embodiment. Amplifier **200** includes three separate inputs **202a, 202b** and **202c**, an output **204**, a first amplifier **236** (also referred to as a first transistor, below), a second amplifier **246** (also referred to as a second transistor, below), a third amplifier **256** (also referred to as a third transistor, below), and an output combiner circuit **240**.

(26) A first signal (RF_in1) received at the first input **202a** is amplified by the first amplifier **236**. A second signal (RF_in2) received at the second input **202b** is amplified by the second amplifier **246**. A third signal (RF_in3) received at the third input **202c** is amplified by the second amplifier **256**.

(27) The first amplifier **236** includes a first transistor (carrier transistor) **238** which may comprise a control terminal coupled to the input **202a**, a source terminal coupled to ground, and a drain terminal as a first amplifier output.

(28) The second amplifier **246** includes a second transistor (first peaking transistor) **248** which may

comprise a control terminal coupled to the second input **202b**, a source terminal coupled to ground, and a drain terminal as a second amplifier output.

(29) The third amplifier **256** includes a third transistor (second peaking transistor) **258** which may comprise a control terminal coupled to the third input **202c**, a source terminal coupled to ground, and a drain terminal as a third amplifier output.

(30) The output combiner circuit **240** is coupled to the first amplifier output and the second amplifier output and the third amplifier output. The output combiner circuit **240** may comprise a first output network **251**, a second output network **252** and a third output network **253**. The first output network **251** may be coupled between the first amplifier output and a final summing node **260**. The second output network **252** may be coupled between the final summing node **260** (and the first amplifier output) and an intermediate summing node **261**. The third output network **253** may be coupled between the third amplifier output and the intermediate summing node **261**.

(31) The first output network **251** comprises a first inductive element **L1** coupled between the first amplifier output and an intermediate node **262**, a first capacitor **C1** coupled between the intermediate node **262** and ground, and a second inductive element **L2** coupled between the intermediate node **262** and the final summing node **260**. The first inductive element **L1** comprises a set of bond wires, and the second inductive element comprises another set of bond wires. The Doherty amplifier **200** comprises integrated passive devices (IPD) (shown in FIG. 3). The first capacitor **C1** is integrally formed in an IPD. The first output network **251** in association with the parasitic output capacitance (C_{ds_c} , FIG. 2B) of the first transistor **236** and a shunted capacitance, **C1**, that is equal to C_{ds_c} are used instead of a quasi quarter-wave transmission line (TL), to provide a 90° phase shift for the output of the first amplifier and the equivalent impedance is **Z1**.

(32) The second output network **252** comprises a third inductive element **L3** coupled between the final summing node **260** and a second intermediate node **263**, a second capacitor **C2** coupled between the second intermediate node **263** and ground, a fourth inductive element **L4** coupled between the intermediate summing node **261** and the second intermediate node **263**. Usually, when the second transistor is larger than the third transistor, then the output parasitic capacitance C_{ds_p1} (FIG. 2B) of the second transistor can be split into two capacitances **C7** and **C8**. **C8** may have a capacitance value chosen to equal the output parasitic capacitance of the third transistor (C_{ds_p2} , FIG. 2B). Then, the capacitance value of **C7** may be chosen to equal capacitance C_{ds_p1} minus capacitance C_{ds_p2} . The second output network **251** in association with the capacitance **C7** and a shunted capacitance **C6** that is chosen to equal to **C7** are used instead of a quasi quarter-wave transmission line (TL), to provide a 90° phase shift for the output of the second amplifier and the equivalent impedance is **Z2**.

(33) The third output network **253** comprises a fifth inductive element **L5** coupled between the intermediate summing node **261** and a third intermediate node **264**, a third capacitor **C3** coupled between the third intermediate node **264** and ground, and a sixth inductive element **L6** coupled between the third intermediate node **264** and the third amplifier output. The third output network **253** in association with the output parasitic capacitance C_{ds_p1} (FIG. 2B) of the second transistor **248** and the output parasitic capacitance C_{ds_p2} (FIG. 2B) of the third transistor **258** are used instead of a quasi quarter-wave transmission line (TL), to provide a 90° phase shift for the output of the third amplifier and the equivalent impedance is **Z3**.

(34) FIG. 2B shows a Doherty amplifier comprising an output combiner using C-L-C-L-C circuits. The Doherty amplifier of FIG. 2B differs from the Doherty amplifier of FIG. 2A, in that the Doherty amplifier of FIG. 2B depicts the parasitic capacitances of the first, second, and third transistors (C_{ds_c} , C_{ds_p1} , and C_{ds_p2}). In addition, the Doherty amplifier of FIG. 2B also includes additional shunt capacitors. For example, shunt capacitor **C5** in the first output network **251** is coupled between summing node **260** and ground. Shunt capacitors **C6** and **C7** are included in the second output network **252**, and they are coupled between ground and summing nodes **260** and **261**, respectively. As shown in FIG. 2B, each of the first output network **251**, the second output

network **252** and the third output network **253** each comprise a C-L-C-L-C circuit, where the output parasitic capacitance of the first transistor (Cds_c), the output parasitic capacitance of the second transistor (Cds_p1) and the output parasitic capacitance of the third transistor (Cds_p2) are included in each C-L-C-L-C network.

(35) FIG. 2C shows a model of the quasi quarter-wave TL formed by a C-L-C-L-C circuit. To form the quasi quarter-wave TL, given C1, C2 and L1, the values of L2, C3 and the characteristic impedance of the quasi quarter-wave TL (Z) could be obtained by the below equations (there are two alternative solutions for L2, C1 and Z1: the equation group (1) is one solution, and the equation group (2) is another).

$$\begin{aligned}
 L_2 &= \frac{1 - \sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1}}{2^{-2}C_2} \\
 C_3 &= \frac{C_1\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (C_1 + 2^{-2}C_1C_2L_1 - 2C_2)}{(^{-2}C_1L_1 - 1)\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (^{-2}C_1L_1 - 1)} \\
 Z &= \frac{-\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (2^{-2}C_2L_1 - 2^{-4}C_1C_2L_1^2 - 1)}{(^{-3}C_1C_2L_1 - C_2)\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (^{-3}C_1C_2L_1 - C_2)} \\
 L_2 &= \frac{1 + \sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1}}{2^{-2}C_2} \\
 C_3 &= \frac{C_1\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (C_1 + 2^{-2}C_1C_2L_1 - 2C_2)}{(^{-2}C_1L_1 - 1)\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} - (^{-2}C_1L_1 - 1)} \\
 Z &= \frac{-\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} + (2^{-2}C_2L_1 - 2^{-4}C_1C_2L_1^2 - 1)}{(^{-3}C_1C_2L_1 - C_2)\sqrt{4^{-4}C_1C_2L_1^2 - 4^{-2}C_2L_1 + 1} + (^{-3}C_1C_2L_1 - C_2)}
 \end{aligned}
 \tag{36} \quad (1) \quad \{ \quad (2)$$

(37) Referring back to FIG. 2B, for simplicity in the real practice, we could let C5 equal to Cds_c, C8 equal to Cds_p1 and C6 equal to C7, and then it results in L1=L2, L3=L4 and L5=L6. In this condition, Given that the Cds of the carrier and peaking transistors are known, the values of C1, C2, C3 and the characteristic impedance Z1, Z2 and Z3 can be adjusted by L1, L3, and L5, respectively.

(38) The impedance at the final summing node **260** is not the real resistance (RL) but a complex one including RL, C5 and C6. Therefore, nine components (L1-L6, C1, C2 and C3) are used to form a 3-way Doherty combiner. The L1, L2, L3, L4, L5 and L6 could be implemented by bond wires, while C1, C2 and C3 can be IPD capacitors (see FIG. 3). Using bond wires as inductors is very low cost and has a very good quality factor and high value of self-resonance.

(39) FIG. 3 is a simplified schematic diagram of an amplifier device **300** in accordance with an embodiment. The packaged amplifier device **300** is a part of a Doherty amplifier.

(40) The device may include a “flat no-leads” device package (e.g., a quad-flat no-leads QFN) or dual-flat no-leads (DFN) device). Such a device includes a “lead frame” consisting of a conductive central flange (“exposed thermal paddle”) and a plurality of perimeter IO pads or leads (referred to as “pin pads” in the application). The flange and the pads are held in fixed orientation with respect to each other (and electrically isolated from each other) with plastic.

(41) The various amplifier dies and the “integrated passive devices” (IPDs) referred to in the application may be all directly attached to the conductive flange, and bond wires may be connected between the top internal surfaces of the pin pads and the dies/IPDs. Plastic molding compound may be then applied over the dies to encapsulate the device. When connected to a PCB, the flange is typically grounded; this provides a ground reference for the encapsulated dies/IPDs. Such a device is typically a surface mount device, so the bottom surfaces of the pin pads are connected (e.g.,

soldered) to corresponding pads on the top surface of the PCB.

(42) An IPD, as discussed herein, is a, typically small, semiconductor die that includes only “passive” components (e.g., capacitors, resistors, inductive elements) integrated therein, as opposed to an “active” device that is a semiconductor die that includes a transistor. In this application, each IPD may include a bond pad on its top surface to which a bond wire may be connected. The IPD includes an integrated, internal capacitor (usually a “metal insulator metal” or MIM capacitor). One terminal of the capacitor is connected to the top bond pad. The other terminal of the capacitor is connected to a conductive layer on the bottom of the IPD. When the IPD is connected (e.g., using solder) to the top surface of the conductive package flange, the second terminal of the capacitor may be grounded.

(43) Referring again to FIG. 3, the physical layout of an amplifier device **300** is illustrated according to one or more embodiments. The packaged amplifier device **300** comprises a first input lead **302a**, a second input lead **302b** and a third input lead **302c**, an output lead **304** on another pin pad, a first amplifier **336**, a second amplifier **346**, a third amplifier **356**, a first inductive element **L1**, a second inductive element **L2**, a third inductive element **L3**, a fourth inductive element **L4**, a fifth inductive element **L5**, a sixth inductive element **L6**, a first capacitor **C1**, a second capacitor **C2**, and a third capacitor **C3**. The various leads **302a**, **302b**, **302c**, **360** each could be a terminal (or pin) of a no-leads package, for example. The **L1**, **L2**, **L3**, **L4**, **L5** and **L6** inductive elements each could be implemented by bond wires. The **C1**, **C2**, and **C3** capacitors each could be implemented as an integrated capacitor within IPDs **371**, **372**, and **373**, respectively. Each of the inductive elements and the capacitors in FIG. 3 correspond to the same-numbered components in FIGS. 2A and 2B.

(44) The first amplifier **336** (e.g., a carrier amplifier of a Doherty amplifier) may include a first transistor **338**. The first transistor **338** comprises a first transistor input coupled to a first input lead **302a** (e.g., through two sets of bond wires and a first input IPD, as shown), and a first transistor output. The second amplifier **346** (e.g., a first peaking amplifier of a Doherty amplifier) comprises a second transistor **348**. The second transistor **348** includes a second transistor input coupled to the second input lead **302b** (e.g., through two sets of bond wires and a second input IPD, as shown), and a second transistor output. The third amplifier **356** (e.g., a second peaking amplifier of a Doherty amplifier) comprises a third transistor **358**. The third transistor **358** includes a third transistor input coupled to the third input lead **302c** (e.g., through two sets of bond wires and a third input IPD, as shown), and a third transistor output.

(45) The first inductive element **L1** is coupled between the first transistor output and a first terminal of the first capacitor **C1**, and the second inductive element **L2** is coupled between the output lead **304** and the first terminal of the first capacitor **C1**. The second terminal of the first capacitor **C1** is coupled to ground.

(46) The third inductive element **L3** is coupled between the output lead **304** and a first terminal of the second capacitor **C2**, and the fourth inductive element **L4** is coupled between the second amplifier output and the first terminal of the second capacitor **C2**. The second terminal of the second capacitor **C2** is coupled to ground.

(47) The fifth inductive element **L5** is coupled between the second amplifier output and the first terminal of the third capacitor **C3**, and the sixth inductive element **L6** is coupled between the second amplifier output and the first terminal of the third capacitor **C3**. The second terminal of the third capacitor **C3** is coupled to ground.

(48) FIG. 4 illustrates a schematic perspective diagram of a packaged amplifier device of FIG. 3 in accordance with an embodiment. In one embodiment, the packaged device is a dual-flat no-leads (DFN) device. All components of the Doherty amplifier **300** are arranged on a common conductive substrate or flange **340**, which may provide a common ground plane for connections to the various grounded capacitances and the source connections of the first, second, and third amplifier transistors. In other embodiments, the packaged device could also be other types device such as quad-flat no-leads (QFN) device.

(49) The packaged amplifier device **400** comprises a first input lead **302a**, a second input lead **302b**, and a third input lead **302c**, an output lead **304**, a first amplifier including a first transistor **338**, a second amplifier including a second transistor **348**, a third amplifier including a third transistor **358**, a first inductive element **L1**, a second inductive element **L2**, a third inductive element **L3**, a fourth inductive element **L4**, a fifth inductive element **L5**, a sixth inductive element **L6**, a first capacitor formed within IPD **371**, a second capacitor formed within IPD **372**, and a third capacitor **C3** formed within IPD **373**.

(50) FIG. 5 is a simplified schematic diagram of a Doherty amplifier **500** with 2nd harmonic control in accordance with an embodiment. Doherty amplifier **500** is identical to Doherty amplifier **200** (FIG. 2A), with one primary difference. The difference between the amplifiers **500**, **200** of FIG. 5 and FIG. 2A is that the Doherty amplifier **500** comprises another inductive element **L13** and capacitance **C7** connected in series between the final summing node **260** and ground. The additional LC circuit comprising the additional inductive element **L13** and capacitance **C7** connected in series between the final summing node **260** and ground may be designed to resonate at two times of the center frequency. Accordingly, the additional LC circuit provides second harmonic control by shunting signal energy at the second harmonic frequency to ground. With this implementation, the second harmonic energy seen by the carrier amplifier is mitigated, and the amplifier performance is within the desirable efficiency area as shown in FIG. 9.

(51) FIG. 6 illustrates the physical layout of an amplifier device of FIG. 5 in accordance with an embodiment. In comparison with FIG. 3, the amplifier device **600** comprises an additional inductive element **L13** and capacitor **C7** connected in series between the output lead **304** and ground. The additional inductive element **L13** (e.g., another set of bondwires) is coupled between the output lead **304** and a first terminal of the capacitor **C7** (e.g., another capacitor integrated within another IPD). The second terminal of the capacitor **C7** is coupled to ground.

(52) FIG. 7 is a Smith chart **700** illustrating second harmonic load-pull data. The drain efficiency is not only determined by the fundamental impedance but also the second harmonic impedance. This data assumes that the fundamental impedance is fixed. For example, the drain efficiency can reach about 78% if the second harmonic impedance is in the good area **710** as shown in FIG. 7. However, the drain efficiency may only be about 60% if the second harmonic impedance is in the bad area **720**. Therefore, the second harmonic impedance may significantly impact the efficiency, and the efficiency difference can reach 18%.

(53) FIG. 8 is a Smith chart **800** illustrating the impedance **S11** presented to the carrier amplifier of the 3-way Doherty amplifier of FIG. 2A without 2nd harmonic control. The impedance includes fundamental impedance (**m2**) and second harmonic impedance (**m8**). As shown in FIG. 8, without second harmonic control, the drain efficiency may be about 70% which is affected by the second harmonic impedance and the load-pull data.

(54) FIG. 9 is a Smith chart **900** illustrating the impedance presented to the carrier amplifier of the 3-way Doherty amplifier of FIG. 5 with 2nd harmonic control. As compared to FIG. 7, the fundamental impedance (**m2**) is fixed. The second harmonic impedance (**m8**) in FIG. 9 is within the good efficiency area. Adding the 2nd harmonic control, the second harmonic impedance (**m8**) will not be changed by any external matching networks connected to the output combiner circuit. The size of the gate periphery may change the impedance seen by the first amplifier.

(55) FIG. 9 is a Smith chart **900** illustrating the impedance presented to the carrier amplifier of the 3-way Doherty amplifier of FIG. 5 with 2nd harmonic control for an embodiment of a Doherty amplifier that uses a gallium nitride (GaN) 1.52 mm die for the carrier amplifier. It should be understood that using a different die size will result in a different impedance, and the impedance will have a different curve from that shown in FIG. 9.

(56) Thus, viewed from one perspective, the present application may enable advantages of high efficiency at deep output back-off (OBO), PCB size reduction, low cost, PCB design friendliness, application flexibility and short go-to-market time. Output Back-Off (OBO) is the power level at

the output of RF amplifier relative to maximum output level possible using the RF amplifier. The deep output back off means that the value of the output back off is larger than the normal back off value. In one embodiment, the value of the normal output back-off is -8 dB and the value of the deep output back-off is -12 dB.

(57) FIG. **10** is a simplified schematic diagram of a 2-way Doherty amplifier **1000** in accordance with another embodiment. Amplifier **1000** includes two separate inputs **1002a**, **1002b** and an output **1004**, a first amplifier **1036**, a second amplifier **1046**, and an output combiner circuit **1040**.

(58) A first signal (RF_in1) received at the first input **1002a** is amplified by the first amplifier **1036**. A second signal (RF_in2) received at the second input **1002b** is amplified by the second amplifier **1046**.

(59) The first amplifier **1036** includes a first transistor (carrier transistor) **1038** which may comprise a control terminal coupled to the input **1002a**, a source terminal coupled to ground, and a drain terminal as a first amplifier output.

(60) The second amplifier **1046** includes a second transistor (first peaking transistor) **1048** which may comprise a control terminal coupled to the second input **1002b**, a source terminal coupled to ground, and a drain terminal as a second amplifier output.

(61) The output combiner circuit **1040** is coupled to the first amplifier output and the second amplifier output. The output combiner circuit **1040** may comprise a first output network **1051** and a second output network **1052**. The first output network **1051** may be coupled between the first amplifier output and a summing node **1060**. The second output network **1052** may be coupled between the second amplifier output and the summing node **1060**.

(62) The first output network **1051** comprises a first inductive element **L1** coupled between the first amplifier output and an intermediate node **1061**, a first capacitor **C1** coupled between the intermediate node **1061** and ground, and a second inductive element **L2** coupled between the intermediate node **1062** and the summing node **1060**. The first inductive element **L1** comprises a set of bond wires, and the second inductive element **L2** comprises another set of bond wires.

(63) The second output network **1052** comprises a third inductive element **L3** coupled between the second amplifier output and a second intermediate node **1062**, a second capacitor **C2** coupled between the second intermediate node **1062** and ground, and a fourth inductive element **L4** coupled between the second intermediate node **1062** and the summing node **1060**. The value of **L3** equals to the value of **L4**.

(64) The first output network imparts a phase delay of a first number of degrees between an input of the first output network and an output of the first output network. The second output network imparts a phase delay of second number of degrees between an input of the second output network and an output of the second output network. The difference between the second number of degrees and the first number of degrees is θ , and θ has a value other than zero.

(65) To get the signal phase aligned at the summing node **1060**, the phase offset θ should be added between the input signals of the first amplifier and the second amplifier. The phase shift between the first amplifier input and the second amplifier input is θ . The analytical expression for θ could be calculated as below:

$$(66) \quad = \pm \cos^{-1} \sqrt{\frac{\alpha^2 - 1}{2 - 1}}, \quad (A2) \text{ or } \pm (\alpha - \cos^{-1} \sqrt{\frac{\alpha^2 - 1}{2 - 1}}) \text{ where } \alpha \text{ is denoted as the ratio of the}$$

peaking amplifier size (or periphery) to the carrier amplifier size, and β is the load modulation range, denoting the ratio of the load impedance seen by the intrinsic current generator of the carrier device at the design output back-off condition to the load impedance seen at the peak power operating.

(67) The output combiner circuit **1040** comprises two CLCL circuits. The first CLCL circuit includes the first output network **1051** and the parasitic capacitor of the first transistor **1038** (**Cds_c**, not illustrated). The second CLCL circuit includes the second output network **1052** and the parasitic capacitor second transistor **1048** (**Cds_p**, not illustrated). In one embodiment, the value of

L1 equals to the value of L2, and the value of L3 equals to the value of L4. Given that the Cds of the carrier and peaking transistors are known, the values of L1, L3, C1, C2 and the complex impedance at the combining node (or admittance $G_L + jY_L$) could be obtained by the equations as below:

$$(68) \quad L_1 = \frac{B_c}{D_c + B_c^2 C_{ds_c} +} \quad (3) \quad L_3 = \frac{B_p}{A_p + B_p^2 C_{ds_p} +} \quad (4)$$

$$C_1 = \frac{(1 - D_c - B_c C_{ds_c})(1 + D_c + B_c C_{ds_c})}{B_c} \quad (5) \quad C_2 = \frac{(1 - A_p - B_p C_{ds_p})(1 + A_p + B_p C_{ds_p})}{B_p} \quad (6)$$

$$G_L = G \quad (7) \quad Y_L = \frac{D_c}{B_c} + \frac{A_p}{B_p} + (C_{ds_c} + C_{ds_p}) - \frac{-1}{(-1)^2 A_p B_p} \quad (8)$$

where the B_c , D_c , A_p , B_p could be obtained by the equations (A5, A6, A7 and A8), and G is a free design variable in the equations (A5, A6, A7 and A8) as described as below.

$$(69) \quad D_c = \pm \sqrt{\frac{(C_{ds_c} - 1)}{(C_{ds_c}^2 - 1)R_0 G}} \quad (A5) \quad B_c = \mp \text{sign}(\cos \theta) \sin \theta \sqrt{\frac{R_0}{G}} \quad (A6)$$

$$B_p = \mp \frac{\sin \theta}{G} \sqrt{\frac{(C_{ds_p} - 1)(C_{ds_p} + 1)R_0}{(-1)^2}} \quad (A7) \quad A_p = \mp \text{sign}(\cos \theta) \sqrt{\frac{R_0}{G}} \cdot \text{Math.} \frac{1}{-1} \quad (A8)$$

(70) The design output back-off (OBO) in dB will be:

$\text{OBO} = 10 \log(\beta(1+\alpha)) \quad (9)$ where α is denoted as the ratio of the peaking amplifier size (or periphery) to the carrier amplifier size, and β is the load modulation range, denoting the ratio of the load impedance seen by the intrinsic current generator of the carrier device at the design output back-off condition to the load impedance seen at the peak power operating.

(71) FIG. 11 illustrates the physical layout of the 2-way amplifier device **1000** of FIG. 10 in accordance with an embodiment. The amplifier device **1100** includes a portion of a Doherty amplifier.

(72) Referring now to FIG. 11, the physical layout of the amplifier device **1000** is illustrated, according to one or more embodiments. The packaged amplifier device **1000** comprises a first input lead **1002a**, a second input lead **1002b**, an output lead **1004**, a first amplifier **1036**, a second amplifier **1046**, a first inductive element **L1**, a second inductive element **L2**, a third inductive element **L3**, a fourth inductive element **L4**, a first capacitor **C1** and a second capacitor **C2**. The **L1**, **L2**, **L3**, and **L4** inductive elements each could be implemented by bond wires. The **C1** and **C2** capacitors each could be implemented as an integrated capacitor within IPDs **1071**, **1072**, respectively.

(73) The first amplifier **1036** may include a first transistor **1038**. The first transistor **1038** comprises a first transistor input coupled to a first input lead **1002a**, and a first transistor output. The second amplifier **1046** comprises a second transistor **1048**. The second transistor **1048** includes a second transistor input coupled to the second input lead **1002b**, and a second transistor output.

(74) The first inductive element **L1** is coupled between the first transistor output and a first terminal of the first capacitor **C1** that is integrated within a first IPD **1071**, and the second inductive element **L2** is coupled between the output lead **904** and the first terminal of the first capacitor **C1**. The second terminal of the first capacitor **C1** is coupled to ground.

(75) The third inductive element **L3** is coupled between the second transistor output and a first terminal of the second capacitor **C2** that is integrated within a second IPD **1072**, and the fourth inductive element **L4** is coupled between the output lead **904** and the first terminal of the second capacitor **C2**. The second terminal of the second capacitor **C2** is coupled to ground.

(76) Referring now to the use of the terms “a” and “an” and “the” and similar referents in the context of describing the subject matter (particularly in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it

were individually recited herein. Furthermore, the foregoing description is for the purpose of illustration only, and not for the purpose of limitation, as the scope of protection sought is defined by the claims as set forth hereinafter together with any equivalents thereof entitled to. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illustrate the subject matter and does not pose a limitation on the scope of the subject matter unless otherwise claimed. The use of the term “based on” and other like phrases indicating a condition for bringing about a result, both in the claims and in the written description, is not intended to foreclose any other conditions that bring about that result. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the application as claimed.

(77) Preferred embodiments are described herein, including the best mode known to the inventor for carrying out the claimed subject matter. Of course, variations of those preferred embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventor expects skilled artisans to employ such variations as appropriate, and the inventor intends for the claimed subject matter to be practiced otherwise than as specifically described herein. Accordingly, this claimed subject matter includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A Doherty amplifier device, comprising: a first input terminal; a second input terminal; a third input terminal; an integrated output combiner circuit comprising a first output network, a second output network, a third output network, an intermediate summing node, and a final summing node; a carrier amplifier including a first amplifier input coupled to the first input terminal, and a first amplifier output coupled through the first output network to the final summing node, wherein the first output network includes a first inductive element in the form of a first set of wirebonds coupled between the first amplifier output and a first intermediate node, a second inductive element in the form of a second set of wirebonds coupled between the first intermediate node and the final summing node, and a first capacitor coupled between the first intermediate node and a ground plane; a first peaking amplifier including a second amplifier input coupled to the second input terminal, and a second amplifier output, wherein the second amplifier output is directly coupled to the intermediate summing node, the second amplifier output and the intermediate summing node are coupled through the second output network to the final summing node, the second output network includes a third inductive element in the form of a third set of wirebonds coupled between the final summing node and a second intermediate node, a fourth inductive element in the form of a fourth set of wirebonds coupled between the second intermediate node and the intermediate summing node, and a second capacitor coupled between the second intermediate node and the ground plane; and a second peaking amplifier including a third amplifier input coupled to the third input terminal, and a third amplifier output coupled through the third output network to the intermediate summing node, wherein the third output network includes a fifth inductive element in the form of a fifth set of wirebonds coupled between the intermediate summing node and a third intermediate node, a sixth inductive element in the form of a sixth set of wirebonds coupled between the third intermediate node and the third amplifier output, and a third capacitor coupled between the third intermediate node and the ground plane, wherein the first output network imparts a phase delay of 90 degrees between an input of the first output network and an output of the first output network and provides a first impedance, the second output network imparts a phase delay of 90 degrees between an input of the second output network and an output of the second output network and provides a second impedance, and the third output network imparts a phase delay of

90 degrees between an input of the third output network and an output of the third output network and provides a third impedance.

2. The device of claim 1, wherein the phase shift between the first amplifier input and the second amplifier input is 0° , and the phase shift between the second amplifier input and the third amplifier input is 90° .
3. The device of claim 1, wherein the final summing node comprises an output terminal of the device.
4. The device of claim 1, further comprising: a conductive substrate, wherein the first, second, and third power transistors and the first, second, and third capacitors are coupled to a surface of the conductive substrate, and the conductive substrate corresponds to the ground plane; and plastic molding compound applied over the first, second, and third power transistors.
5. The device of claim 1, wherein the device is a packaged amplifier device that has a device type selected from a group consisting of a dual-flat no-leads device and a quad-flat no-leads device.
6. The device of claim 1, wherein the first capacitor is integrated within a first integrated passive device (IPD), the second capacitor is integrated within a second IPD, and the third capacitor is integrated within a third IPD.
7. The device of claim 1, wherein: the device is a packaged amplifier device; the first, second, and third terminals are first, second, and third leads of the packaged amplifier device; the final summing node comprises an output terminal of the device; and the output terminal corresponds to a fourth lead of the packaged amplifier device.
8. The device of claim 1, further comprising a resonator coupled between the final summing node and the ground plane, wherein the resonator comprises a seventh inductive element and a fourth capacitor coupled in series, and the resonator is configured to resonate at a second harmonic frequency of a center frequency of operation of the device.
9. The device of claim 1, wherein: the carrier amplifier comprises a first input impedance matching network and a first power transistor, wherein the first input impedance matching network is configured to match an impedance between the first input terminal and the first power transistor; the first peaking amplifier comprises a second input impedance matching network and a second power transistor, wherein the second input impedance matching network is configured to match an impedance between the second input terminal and the second power transistor; and the second peaking amplifier comprises a third input impedance matching network and a third power transistor, wherein the third input impedance matching network is configured to match an impedance between the third input terminal and the third power transistor.
10. The device of claim 9, wherein: the first power transistor is a first field effect transistor with a first gate terminal coupled to the first input impedance matching network, a first drain terminal coupled to the first inductive element, and a first source terminal coupled to the ground plane, the second power transistor is a second field effect transistor with a second gate terminal coupled to the second input impedance matching network, a second drain terminal coupled to the intermediate summing node, and a second source terminal coupled to the ground plane, and the third power transistor is a third field effect transistor with a third gate terminal coupled to the third input impedance matching network, a third drain terminal coupled to the sixth inductive element, and a third source terminal coupled to the ground plane.
11. A Doherty amplifier, comprising: a first input; a second input; a third input; an output; an output combiner circuit with first, second, and third combining network inputs, first, second, and third output networks, an intermediate summing node corresponding to the second combining network input, a final summing node, and a combining network output corresponding to the final summing node and coupled to the output directly; a carrier amplifier including a first amplifier input coupled to the first input, and a first amplifier output coupled to the first combining network input; a first peaking amplifier including a second amplifier input coupled to the second input, and a second amplifier output directly coupled to the second combining network input; and a second peaking

amplifier including a third amplifier input coupled to the third input, and a third amplifier output coupled to the third combining network input, wherein the first output network includes a first inductive element in the form of a first set of wirebonds coupled between the first combining network input and a first intermediate node, a second inductive element in the form of a second set of wirebonds coupled between the first intermediate node and the combining network output, and a first capacitor coupled between the first intermediate node and a ground plane, wherein the second output network includes a third inductive element in the form of a third set of wirebonds coupled between the combining network output and a second intermediate node, a fourth inductive element in the form of a fourth set of wirebonds coupled between the second intermediate node and the second combining network input, and a second capacitor coupled between the second intermediate node and the ground plane, wherein the third output network includes a fifth inductive element in the form of a fifth set of wirebonds coupled between the second combining network input and a third intermediate node, a sixth inductive element in the form of a sixth set of wirebonds coupled between the third intermediate node and the third combining network input, and a third capacitor coupled between the third intermediate node and the ground plane, and wherein the first output network imparts a phase delay of 90 degrees between an input of the first output network and an output of the first output network and provides a first impedance, the second output network imparts a phase delay of 90 degrees between an input of the second output network and an output of the second output network and provides a second impedance, and the third output network imparts a phase delay of 90 degrees between an input of the third output network and an output of the third output network and provides a third impedance.

12. The Doherty amplifier of claim 11, wherein; the carrier amplifier comprises a first input impedance matching network and a first power transistor, wherein the first input impedance matching network is configured to match an impedance between the first input and the first power transistor; the first peaking amplifier comprises a second input impedance matching network and a second power transistor, wherein the second input impedance matching network is configured to match an impedance between the second input and the second power transistor; and the second peaking amplifier comprises a third input impedance matching network and a third power transistor, wherein the third input impedance matching network is configured to match an impedance between the third input and the third power transistor.

13. The Doherty amplifier of claim 11, wherein the phase shift between the first amplifier input and the second amplifier input is 0° , and the phase shift between the second amplifier input and the third amplifier input is 90° .

14. The Doherty amplifier of claim 11, wherein the final summing node comprises the output of the Doherty amplifier.

15. The device of claim 12, further comprising: a conductive substrate, wherein the carrier and peaking amplifiers and the first and second capacitors are coupled to a surface of the conductive substrate, and the conductive substrate corresponds to the ground plane; and plastic molding compound applied over the carrier and peaking amplifiers.

16. The device of claim 12, wherein the device is a packaged amplifier device that has a device type selected from a group consisting of a dual-flat no-leads device and a quad-flat no-leads device.

17. The device of claim 12, wherein the device is a packaged amplifier device, and the first and second terminals are first and second leads of the packaged amplifier device.

18. The device of claim 17, wherein the output terminal corresponds to a summing node and to a fourth lead of the packaged amplifier device.

19. A Doherty amplifier device, comprising: a first input terminal; a second input terminal; an output terminal; an integrated output combiner circuit comprising a first output network and a second output network; a carrier amplifier including a first amplifier input coupled to the first input terminal, and a first amplifier output coupled through the first output network to the output terminal, wherein the first output network includes a first inductive element in the form of a first set

of wirebonds coupled between the first amplifier output and a first intermediate node, a second inductive element in the form of a second set of wirebonds coupled between the first intermediate node and the output terminal, and a first capacitor coupled between the first intermediate node and a ground plane; and a peaking amplifier including a second amplifier input coupled to the second input terminal, and a second amplifier output coupled through the second output network to the output terminal, wherein the second output network includes a third inductive element in the form of a third set of wirebonds coupled between the second amplifier output and a second intermediate node, a fourth inductive element in the form of a fourth set of wirebonds coupled between the second intermediate node and the output terminal, and a second capacitor coupled between the second intermediate node and the ground plane, wherein the first capacitor is integrated within a first integrated passive device (IPD), wherein the second capacitor is integrated within a second IPD, and wherein the device is a packaged amplifier device, and the first input terminal, the second input terminal, and the output terminal are first, second, and third leads of the packaged amplifier device, and the lead corresponds to a summing node for signals produced by the first and second output networks.

20. The Doherty amplifier device of claim 19, wherein a phase shift between the first amplifier input and the second amplifier input is θ , wherein the first output network imparts a phase delay of a first number of degrees between an input of the first output network and an output of the first output network, wherein the second output network imparts a phase delay of a second number of degrees between an input of the second output network and an output of the second output network, and wherein the difference between the second number of degrees and the first number of degrees is θ , and θ is a number of degrees other than zero.
